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The Editor-in-Chief
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Dear Editor,

I submit the manuscript “Coupled Condensation–Melting Heat Transfer and Thermal Design Limits of a Steam-Assisted Pipeline Cleaning Crawler: A Probabilistic Modeling Study” as a research article for consideration in Applied Thermal Engineering.

When steam is used to melt a deposit from a pipe wall, the heat reaches the solid through a condensate film and through any molten deposit that stays on the surface, and the same steam attacks the insulation that protects the equipment doing the work. The manuscript treats this coupled condensation –melting problem through one application, a tracked crawler for steam-assisted cleaning of hydrocarbon pipelines, with a probabilistic model that couples transient conduction through an aluminum, silica-aerogel and titanium wall to a lumped battery pack, film condensation, a melt model with a retained liquid film, an orifice model of the pressure relief, and a steam budget.

The manuscript makes five contributions to the thermal-engineering literature. (i) The retained film is expressed through a dimensionless resistance $\Pi_f = h\,s / k_l$, so that the heat flux is $q_0/(1 + \Pi_f)$, and is predicted by a dynamic balance of melt production and gravity drainage instead of being prescribed: the median range per deployment is 138 m if the melt leaves at once, 18 m with the dynamic film (0.35 mm) and 2.3 m if none drains, and a range of 100 m needs Π_f below about 0.3. (ii) The quasi-steady melt formulas and the conduction solver are benchmarked against an independent enthalpy-method solver, the exact Neumann solution and an independent stiff-ODE integration (errors below 5% for instant removal; the no-drainage formula is 18% too fast), and the three condensation representations compared; the coupled model could not be validated, because no experiment was identified, and the manuscript says so. (iii) Global (Sobol) sensitivity and correlated-input analyses, and a separation of the spread caused by model form (a factor of 59 in the median range) from the spread caused by parameters within a model. (iv) A constrained design search and Pareto set, and a pressure–temperature check that shows the first design's steam at 198 °C exceeding a Class 150 rating, with a revised envelope (192 °C) that costs 3% of the range. (v) Deployment limits quantified, with the dependence on melt removal stated: a 1 km line needs a median of 57 deployments with the dynamic film and 8 if the melt leaves at once (no 45 min deployment reaches 300 m at the setpoint in either case), the hull does not fit pipes below 8 in, the pack probability falls from 98.9% at 25 °C to 22% at 45 °C, and heating the pipe wall adds 1.8 times the melting steam.

The study is explicit about what it is. Every input distribution is an assumed design-stage value, because no published distribution of these deposits, steam conditions or tool was found; the numerical implementation is verified (the first suite is retained unchanged, every added mechanism reduces to it when switched off, and 62 automated checks pass) but the coupled model is not validated physically; and the success percentages are model probabilities under the assumed distributions, not field reliabilities. The conclusions rank the measurements that would matter most, beginning with the retained melt film on steel.

The code and results are openly available at https://github.com/sandlerleon/tamprc-pipeline-crawler-mc and archived at https://doi.org/10.5281/zenodo.24045599 (code, all versions); the manuscript and its Supplementary Information are archived at https://doi.org/10.5281/zenodo.24045595 (all versions). One command regenerates every number and figure.

Disclosure. The application was taken from a public industrial scouting challenge on a third-party innovation platform; I have no financial relationship with the seeker or the platform, and the work received no external funding.

Author affiliation and email address. I am an independent researcher with no current institutional affiliation, and I am the sole author. The corresponding-author email address, sandler.leon@gmail.com, is my personal address.

A generative AI tool (Claude, Anthropic) assisted with code development, numerical experiments, drafting and editing; I reviewed and checked all of it and take full responsibility for the content, as disclosed in the manuscript. The manuscript is original, has not been published elsewhere and is not under consideration at any other journal. I am the sole author and have no competing interests.

Thank you for considering this submission.

Sincerely,

Leon Sandler